

Case Study Report: Software Project Plan

Preparation of Scope, Timeline, and Team Roles (Sample App: E-Learning Platform)

1. Aim and Objectives

Aim

The aim of this case study is to prepare a comprehensive software project plan detailing the scope, project timeline, and team roles for a sample real-world application, an online E-Learning Platform.

Objectives

- Define the project scope to establish clear boundaries for the application's features.
- Allocate team roles and define individual responsibilities for smooth execution.
- Create a structured timeline (project schedule) with distinct phases and milestones.
- Demonstrate how a formal project plan ensures resource efficiency and timely delivery.

2. Project Overview

The selected sample application is an **E-Learning Platform** designed to connect students with educational courses. It requires a stable back-end to handle user data, a front-end interface for video streaming, and a secure payment gateway for course purchases. A robust project plan is required to guide the development team from initiation to deployment.

3. Project Scope

In-Scope Features

- **User Management:** Registration, secure login, and profile management for Students and Instructors.
- **Course Management:** Uploading video lectures, creating quizzes, and managing course content.
- **Learning Interface:** A responsive video player with progress tracking and course completion certificates.
- **Payment Integration:** Secure processing of course purchases and instructor payouts.
- **Admin Dashboard:** User moderation, revenue tracking, and system analytics.

Out-of-Scope Features (For Future Releases)

- Live 1-on-1 video tutoring sessions.
- Offline desktop application.
- Physical textbook ordering and delivery.

4. Team Roles and Responsibilities

To successfully deliver the platform, the following key personnel are assigned to the project team:

Role	Primary Responsibilities
Project Manager (PM)	Oversees the entire project, manages risks, tracks the timeline, and ensures team communication.
Business Analyst (BA)	Gathers detailed software requirements, documents the scope, and acts as a bridge between stakeholders and developers.
UI/UX Designer	Designs the user interface, creates wireframes, and ensures a smooth and intuitive user experience for students.
Frontend Developer	Builds the visible parts of the application (video player, dashboards) using frameworks like React or Angular.
Backend Developer	Develops the server-side logic, manages the database, and integrates the payment gateway using Node.js or Python.
QA Engineer	Writes and executes test cases to identify bugs, checks system security, and ensures the app meets quality standards.
DevOps Engineer	Manages cloud server deployment, automates updates (CI/CD), and monitors system uptime.

5. Project Timeline and Milestones

The project follows a standard 14-week Agile-based development cycle. The timeline is divided into distinct phases to ensure steady progress and continuous testing.

Phase	Activities	Duration	Milestone / Deliverable
Phase 1: Initiation & Planning	Gather requirements, finalize scope, and allocate resources.	Weeks 1 - 2	Approved Project Plan & SRS Document
Phase 2: UI/UX Design	Create wireframes, mockups, and final interface designs.	Weeks 3 - 4	Approved Design Prototypes
Phase 3: Development	Write frontend and backend code, integrate APIs, build database.	Weeks 5 - 10	Completed Source Code (Beta Version)
Phase 4: Quality Assurance (QA)	Perform unit, integration, and security testing. Fix bugs.	Weeks 11 - 12	Bug-Free Stable Application
Phase 5: Deployment	Set up cloud infrastructure, release to production environment.	Week 13	Live Application Launch
Phase 6: Maintenance	Monitor system performance, gather user feedback, patch issues.	Week 14 Onwards	Ongoing Support and Updates

6. Risk Management Plan

A critical part of the project plan is identifying potential risks and defining mitigation strategies.

- **Risk: Schedule Delay.** *Mitigation:* Adopt Agile sprints to ensure continuous delivery and prioritize core features first.

- **Risk: Scope Creep (Adding too many features).** *Mitigation:* Strictly enforce the In-Scope boundaries defined in Phase 1. Any new feature requests will be pushed to a future release version.
- **Risk: Payment Gateway Failure.** *Mitigation:* Use a highly reliable third-party provider (e.g., Stripe, PayPal) and perform rigorous security testing before launch.

7. Results

The preparation of this software project plan resulted in a clearly defined roadmap for the E-Learning Platform. By outlining the scope, assigning team roles, and establishing a structured timeline, the project team successfully eliminated ambiguities early in the process. The expected outcomes of following this plan include seamless communication across departments, prevention of unapproved scope additions, and successful on-time deployment within the designated 14-week target.

8. Conclusion

A well-prepared software project plan is the foundation of successful application development. By clearly defining the project scope, the team avoids unnecessary work and stays focused on the core objectives. Assigning specific roles ensures accountability, while a structured timeline keeps the project on track for a timely launch. For this E-Learning Platform, adhering to this comprehensive 8-step project plan will guarantee a high-quality, secure, and user-friendly final product delivered within budget and schedule.